Debottam Dutta

Room 261, Coordinated Science Laboratory, Urbana, IL, 61801

dd24@illinois.edu | debottam-dutta7.github.io | Google Scholar | debottam-dutta | debottam-dutta7

RESEARCH INTERESTS

Compositional Capabilities of Generative Models (Diffusion, Flows, EBMs); User Preference Alignment; Audio and Music Generation; Multi-channel Source Separation; Efficient and Robust Inference.

EDUCATION

Ph.D. in Electrical and Computer Engineering
University of Illinois Urbana-Champaign — Aug, 2022–Present
Advisor: Prof. Romit Roy Choudhury (SiNRG)

M.Tech in Signal Processing
Indian Institute of Science, Bangalore — July, 2021
Advisor: Prof. Sriram Ganapathy (LEAP Lab)

B.Tech in Electronics and Communication Engineering
National Institute of Technology, Silchar — May, 2018

RESEARCH EXPERIENCE

Graduate Research Assistant — *SiNRG Lab, University of Illinois Urbana-Champaign* 2022–Present
Advisor: Prof. Romit Roy Choudhury

- **Generative Models:** Improving generation quality using efficient samplers; personalized image generation; accelerating generation at inference time using latent caching.
- **Speech and Audio Enhancement for AR Glasses:** Real-time speech enhancement for AR glasses using deep learning and signal processing techniques.

Student Researcher — *LEAP Lab, Indian Institute of Science* 2020–2022
Advisor: Prof. Sriram Ganapathy

- **Acoustics for Digital Health:** Designed interpretable frameworks for COVID-19 detection using acoustic signals (cough, breath, speech) as primary biomarkers.

PREPRINTS

[1] **arXiv:** Debottam Dutta, Jianchong Chen, Jaehoon Hahm, Romit Roy Choudhury, "TILT: Model-Intrinsic Reward Alignment for Compositional Diffusion" (Paper)

[2] **arXiv:** Debottam Dutta, Chaitanya Amballa, Zhongweiyang Xu, Yu-Lin Wei, Romit Roy Choudhury, "Learning Energy-based Variational Latent Prior for VAEs" (Paper)

PUBLICATIONS

[3] **ICML 2026:** Rajalaxmi Rajagopalan, Debottam Dutta, Yu-Lin Wei, Romit Roy Choudhury, "Personalized Image Generation via Human-in-the-loop Bayesian Optimization" (Paper)

[4] **ICML 2026 Workshop:** Debottam Dutta, Jaehoon Hahm, Jianchong Chen, Romit Roy Choudhury, "Can Intrinsic Rewards Enable Compositional Alignment in Diffusion?" (Paper)

[5] **ICLR 2026:** Debottam Dutta, Jianchong Chen, Rajalaxmi Rajagopalan, Yu-Lin Wei, Romit Roy Choudhury, "Steer Away from Mode Collisions: Improving Composition in Diffusion" (Paper)

[6] **HotMobile 2026:** Zhijian Yang, Sattwik Basu, Chaitanya Amballa, Debottam Dutta, Srihari Nelakuditi, Romit Roy Choudhury, "Inferring Indoor Layouts using Audio" (Paper)

[7] **ICASSP 2025:** Sattwik Basu, Debottam Dutta, Yu-Lin Wei, Romit Roy Choudhury, "Estimating Multi-chirp Parameters using Curvature-guided Langevin Monte Carlo" (Paper)

[8] **NeurIPS 2024 Workshop:** Zhongweiyang Xu, Debottam Dutta, Yu-Lin Wei, Romit Roy Choudhury, "Multi-Source Music Generation with Latent Diffusion" (Paper)

[9] **Interspeech 2022:** Debottam Dutta, Debarpan Bhattacharya, Sriram Ganapathy, et al., "Acoustic Representation Learning on Breathing and Speech Signals for COVID-19 Detection" (Paper)

Full list on Google Scholar.

PROJECTS

Diagnosing and Improving Compositional Generation in Diffusion (Paper 1) (Paper 2) (Demo)

- *Why does composition fail?* Proposed the "mode-overlap hypothesis": text-to-image diffusion models drop or overshadow concepts because the multi-concept distribution overlaps with single-concept ones, which pulls sampling toward a dominant concept.
- *Can sampling avoid it without retraining?* Proposed CO3, a training-free corrector that steers sampling away from overlap regions and toward modes where all concepts are present (ICLR 2026).
- *Can a model judge its own failures?* Proposed TILT, which treats composition as alignment to a reward taken from the model itself. The reward is defined purely in distribution space and is thus modality-agnostic: the same reward works for both image and audio, removing the need for external reward models (arXiv).

Controllable Music Generation using Latent Diffusion (Paper) (Demo)

- Designed a Multi-Source Latent Diffusion Model (MSLDM) to jointly model synchronized music stems and enable controllable generation.
- Achieved best FAD and human evaluation scores in the controllable music generation task.

Real-time Speech Enhancement for AR Glasses (Report)

- Proposed a speaker separation and enhancement framework for AR glasses with six microphones, using a combination of deep learning and beamforming techniques.
- Met the real-time latency constraint of 50 ms and achieved *second place* in the SPEAR Challenge.

HONORS AND AWARDS

- **Robert T. Chien Memorial Award** (ECE, UIUC).
- **AR Buck "Knight" Fellowship** (ECE, UIUC).
- **AICTE-PG Scholarship** (Govt. of India).
- **Ishan-Uday Scholarship** (UGC, India).
- **Ananda Ram Borooah Award** (Govt. of Assam, India).

WORK EXPERIENCE

Research Fellow — LEAP Lab, IISc, Bangalore May 2021 – June 2022

- Contributed to data collection and curation for Coswara (Dataset) (Web App), a large-scale respiratory sound dataset (cough/breath/speech) for COVID-19 screening, now widely used in digital health research.
- Built deep learning pipelines for diagnostic audio classification.

TEACHING EXPERIENCE

Teaching Assistant — Dept. of ECE, University of Illinois Urbana-Champaign

- *Global Education Training (GET): Generative AI Algorithms* Summer'26
- *ECE 498/598: Deep Generative Models* Fall'25, Spring'26
- *ECE 434: Real-World Algorithms for IoT and Data Science* Spring'24
- Duties: office hours, grading and class logistics.

Teaching Assistant — Dept. of EE, Indian Institute of Science, Bangalore Spring'21

- *E9 205: Machine Learning for Signal Processing*; office hours, grading and class logistics.

SKILLS

Programming: Python, C/C++, MATLAB, Shell
Tools: Git, LaTeX, Slurm, Weights&Biases
ML/AI Frameworks: PyTorch, JAX, CUDA, Hugging Face Diffusers/Transformers

SERVICE

- Co-organizer, Second DiCOVA Challenge (Website) Aug–Oct, 2021
- Reviewer for ICASSP 2025, AAAI 2025, ICML 2026, NeurIPS 2026, AAAI 2026, ICLR 2027.